

BHAGYA M. SUBRAYAN

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Steward Observatory, 933 N Cherry Ave, Tucson, AZ, 85712

RESEARCH INTERESTS

- Multi-wavelength observations of astrophysical transients in time-domain astronomy
- Dust reservoirs and mass-loss mechanisms in massive stars
- Early photometric and spectroscopic observations of young transients
- Electromagnetic counterparts of gravitational wave (GW) events
- Maximizing scientific return from all-sky surveys with state-of-the-art technologies

CURRENT POSITION

Postdoctoral Research Associate

Sept. 2024 – Present

Steward Observatory, University of Arizona, Tucson, AZ, USA

Advisor: Prof. David J. Sand

EDUCATION

Ph. D. Physics and Astronomy

Aug. 2018 – 2024

Purdue University, West Lafayette, Indiana, USA

Thesis Title: Constraining Explosion Physics and Progenitors of Transients via Statistical Inferencing of All Sky Survey Data Streams Advisor: Prof. Danny Milisavljevic

M. S. Physics, Major in Physics and Minor in Chemistry

Aug. 2013 – May 2018

Indian Institute of Science Education and Research (IISER), India

Integrated B. S - M. S Program, CGPA: 8.59 / 10

HONORS AND AWARDS

- George W. Tautfest Award for Outstanding Graduate Student in High Energy, Nuclear or Astrophysics, Purdue April 2024
- Karl Lark-Horvitz Award for Outstanding Research Accomplishments, Purdue April 2024
- Charlotte Ida Litman Tubis Award for Excellence in Scientific Communication, Purdue April 2024
- Lijuan Wang Memorial Award, Physics and Astronomy, Purdue April 2021, April 2023
- Women in Science Program (WISP) Travel Grant, Purdue Fall 2022
- Summer Research Grant, Physics and Astronomy, Purdue July 2020 – August 2020

PUBLICATIONS

Authored 35+ publications (500+ citations), including 4 first-author papers (70+ citations), 1 (in prep)

First-author:

5. **Subrayan, B.**, Sand, D., Culbert, O., 2026. A JWST/MIRI Study of Dust in a Sample of Normal Core Collapse Supernovae (submitted: [arXiv](#))
4. **Subrayan, B.**, Milisavljevic, D., Garretson, B. et al 2026. Insights from Late Time and HST Observations of the Extreme Nuclear Transient AT2021lwx (*submission in Fall 2026*)
3. **Subrayan, B.**, Sand, D., Bostroem, K. A., et al. 2025. Early Shock Cooling Observations and Progenitor Constraints of Type IIb Supernova SN 2024uwq, ApJL, 990, L68
2. **Subrayan, B.**, Milisavljevic, D., Chornock, R., et al., 2023. Scary Barbie: An Extremely Energetic, Long-duration Tidal Disruption Event Candidate without a Detected Host Galaxy at $z = 0.995$, ApJL, 914, L9
1. **Subrayan, B.**, Milisavljevic, D., Moriya, T. J., et al. 2023. Inferencing Progenitor and Explosion Properties of Evolving Core-collapse Supernovae from Zwicky Transient Facility Light Curves, ApJ, 945, 4

Major Contributions as Co-author:

6. Sand, D. J., **Subrayan, B.**, Ransome, C., et al., 2026. Nearby Supernova Science in the Era of High Cadence, Deep Time Domain Surveys, [arXiv](#)
5. Ransome, C., **Subrayan, B.**, Sand, D. J., 2026. First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532, [arXiv](#)
4. Vieira, N., Franz, N., **Subrayan, B.** et al. 2026. Search For a Counterpart to the Subsolar Mass Gravitational Wave Candidate S251112cm, arXiv:2603.17009
3. Franz, N., **Subrayan, B.**, Kilpatrick, C. D., et al. 2025. Optimizing Kilonova Searches: A Case Study of the Type I Ib SN 2025ulz in the Localization Volume of the Low-significance Gravitational Wave Event S250818k, ApJL, 994, L45
2. Pearson, J., **Subrayan, B.**, Sand, D., et al. 2025. Mid-infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw, ApJ, 993, 213
1. Moriya, T. J., **Subrayan, B.**, Milisavljevic, D., Blinnikov, S., I. 2023. Synthetic red supergiant explosion model grid for systematic characterization of Type II supernovae, PASJ, psad024

16 additional contributions as co-author. 80 Transient Name Server (TNS) reports, Astronotes, and circulars in total.

TELESCOPE PROPOSALS AND SUPPORT GRANTS

Summary: PI of 11 programs with **\$181,103** in associated support grants. Cumulative time awarded as PI/CoI/team member: JWST >20.8 hr (4 programs); HST 5 programs; Keck 5 nights; Gemini >200 hr; Magellan 4 nights; MMT >10 nights; Bok 15 nights; CTIO/DECam 77.1 nights; SOAR >60 nights; LCO >100 nights.

As Principal Investigator (PI)

- JWST - *Resolving the Host Galaxy of AT 2021lwx with JWST: Testing SMBH Host Galaxy Scaling Relations with an Extreme Nuclear Transient (ENT):* GO-11424, Cycle 5, **7.6 hours, \$114,455**
- HST - *HST Observations of the Most Energetic and Luminous Optical Transient AT 2021lwx aka Scary Barbie:* GO-17748, Cycle 32, **3 orbits, \$53,283**
- Keck + Gemini - *Search for Late-Time Signatures of Circumstellar Medium Interaction and Dust Formation in Normal Core-Collapse Supernovae:* Keck 2026B-N172 (NASA/Keck), **0.5 night, \$13,365**; Gemini 2026A, **8.25 hours**
- Magellan + MMT - *Follow-up of Electromagnetic Counterparts to Gravitational Wave Events in O4 and O4-IR1:* Steward/University of Arizona, 2025B–2026B, **4 nights** Magellan + **4 nights** MMT
- Bok - *Chasing Young Supernovae: A Bok/90prime + ZTF Pilot to Prepare for LSST's Early Transient Science:* 2025B, **6 nights**
- Bok - *Bok Telescope Pre-Surveys for ULTRASAT: Laying the Groundwork for UV Transient Science:* 2025A, **9 nights**
- LCO - *Testing Science-driven Prioritization of Transients with LCO for Large-Scale Surveys:* 2021A, **5 hours**

As Co-Investigator (co-I)

- JWST - *Is There Enough? Cosmic Dust Formation in Normal Core-Collapse Supernovae in the First ~5 Years Post-Explosion*: GO-3295, Cycle 2 (PI: D. Sand); **primary analyst**
- JWST - *Watching Dust Formation in Real Time in Two Very Nearby Core Collapse Supernovae*: GO-7881/GO-12572, Cycles 4–5, **10.3 hours**; **primary analyst**
- JWST - *NIR+MIR Spectroscopy of the Nearby Broad Line Type Ic SN 2024abup: r-process, Dust and Explosion Physics*: DD-6803, Cycle 3, **2.9 hours**
- HST - *Bridging the Gap Between Strong and Weak Mass Loss with Nebular UV Spectra of SN 2025tis and SN 2026ayt*: GO-18394
- HST - *Unprecedented Early UV Spectroscopy of a Nearby Supernova*: GO-18368
- HST - *Echoes of Dying Breaths: Measuring Mass Loss with Late-time Observations of SN 2023ixf and SN 2024ggi*: GO-18349
- HST - *Completing a Legacy Dataset with Deep HST Imaging of 121 Nearby Star-Forming Galaxies*: GO-18338
- CTIO/DECam + SOAR - *DECam Shadowing of LSST to Discover the Youngest Transients in the Nearby Universe*: 2025B–2028A, **77.1 nights** DECam + **3.7 nights** SOAR
- SOAR - *The Public AEON Spectroscopic Survey for Transient Astronomy (PASSTA)*: 2025B–2028A, **60 nights**
- Ground-based - CoI on numerous telescope proposals including Gemini (22 programs, 2023–2026), Keck, IRTF, MMT (17 programs via AZTEC), Magellan, SOAR, Las Cumbres Observatory, and CTIO/DECam

CONFERENCE/MEETING TALKS

Invited Talks

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| • Departmental Talk, Dunlap Institute of Astronomy and Astrophysics | Toronto, August 2026 |
| • ASU Spring 2026 Workshop on MMA, Arizona State University | Arizona, May 2026 |
| • Talk to Steward Observatory Stakeholders, University of Arizona | Arizona, November 2025 |
| • Observational Astronomy Meetings, CIERA, Northwestern University | Illinois, January 2024 |
| • The High-Energy Astrophysics Division (HEAD) Frontier Seminar Series | Online, December 2023 |
| • Transient Science @ Space Telescope (TSST) Talk, STScI | Maryland, November 2023 |
| • AstroCoffee, Johns Hopkins University | Maryland, November 2023 |
| • Transient Group Meeting, Harvard CfA | Massachusetts, November 2023 |
| • Monday Afternoon Talk (MAT), MIT Kavli Institute | Massachusetts, November 2023 |
| • BigBoom Talk Series, University of Arizona | Online, October 2023 |

Contributed Talks

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| • LSST Discovery Alliance Catalyst Symposium | August 2026 |
| • RAPID: Hot-wiring Conference, Caltech (<i>Awarded Travel Support</i>) | California, March 2026 |
| • MMA in the Dynamic Universe, Kyoto University (<i>Awarded Travel Support</i>) | Japan, February 2026 |
| • 247th American Astronomical Society (AAS) Winter Meeting | Phoenix, January 2026 |
| • Transients from Space, STScI | Baltimore, March 2025 |

- Rise_Time Conference, Purdue University Indiana, August 2024
- Rare Gems in Big Data Conference, NOIRLab Arizona, May 2024
- 243rd American Astronomical Society (AAS) Winter Meeting New Orleans, January 2024
- BOOM! An LSST Workshop Illinois, July 2022
- Spoken-WERDD Symposium Online, November 2021

TIME-DOMAIN ASTRONOMY COLLABORATIONS

- Distance less than 40 Mpc (DLT40)
- SHADOW (DECam Shadowing of LSST to Discover the Youngest Transients in the Nearby Universe)
- The Public AEON Spectroscopic Survey for Transient Astronomy (PASSTA)
- Arizona Transient Exploration and Characterization (AZTEC)
- Recommender Engine For Intelligent Transient Tracking (REFITT)
- LSST Discovery Alliance Transient and Variable Stars Science (TVS), TVS-LVK Coordination
- Searches After Gravitational waves Using Arizona Observatories (SAGUARO)
- Tool for Rapid Object Vetting and Examination (TROVE)
- University of Arizona new 90" instrument on Bok Telescope (Instrumentation and Commissioning)
- ULTRASAT Collaboration
- Global Supernova Project (GSP)

TECHNICAL EXPERIENCE

Technical Expertise: Reduction pipeline development (Imaging/Spectroscopy), Statistical Data Analysis, Machine Learning, High Performance Computing, Integrated Field Unit, Optical Image Processing, Optical Spectroscopy, Data Mining and Wrangling, Python, C , C++, SQL, HTML, IDL
Software: JWST Reduction Pipelines, Django-DB, spacephot, JHAT, PyRAF, IRAF, ds9, LaTeX, TensorFlow, PyTorch, ccdproc, scikit-learn, GitHub, emcee, dynesty, Astropy, MOSFiT, SuperBol, Jupyter Lab, Matplotlib, Seaborn, Microsoft Applications

TEACHING AND MENTORING

- Pima Spring School Guest Lecturer, Pima School, AZ Spring 2025
- Guest Lecture: AST 567 - Observational Techniques in Astronomy, Purdue, IN Fall 2023
- Guest Lecture: AST 364 - Stars and Galaxies, Purdue, IN Spring 2023
- Olivia Culbert (UG), University of Arizona 2024-2026
Project: Dust in Normal Core-collapse Supernovae (Incoming Master's Student at San Diego State University) - Finalist at Chambliss Competition at 247 AAS Meeting, Phoenix, AZ, 2026
- Braden Garretson (UG), Purdue University 2020 - 2024
Project: Machine Learning Frameworks for Rapid Inference of Light Curves from All-Sky Surveys.
- Moira Andrews (UG) (current graduate student at University of California, Santa Barbara) 2022
Provided guidance to revise application materials for graduate school.

LEADERSHIP, ACADEMIC SERVICE AND OUTREACH

Leadership

- **Women in Science Program (WISP), College of Science, Purdue University** Summer 2020–2023
 - Designed and organized a monthly seminar series featuring invited speakers on professional development and scientific topics. Supported women and non-binary STEM graduate students through community-building and career development initiatives.

- **Spoken-WERDD Symposium Scientific Organizing Committee (SOC)** 2022
 - Served on the Scientific Organizing Committee. Coordinated abstract collection, talk scheduling, and development of the symposium code of conduct.
- **PASSTA Supernova Advocacy Chair** 2025B–2026A
 - Led the PASSTA SN Advocacy group, coordinating follow-up and allocating survey resources to obtain UVOIR photometric and spectroscopic observations of nearby supernovae.

Academic Service

- **Journal Referee** 2024-2026
 - Referee for *Nature Astronomy*, *Monthly Notices of the Royal Astronomical Society (MNRAS)*, and *Astronomy & Astrophysics (A&A)*.
- **Proposal Reviewer**
 - Panel reviewer for *HST* (Cycles 33–34), *JWST* Director’s Discretionary Time (DDT), NASA panels

Outreach and Community Engagement

- **National Public Radio (NPR) 1A Show – Guest Speaker** 2023
 - Invited to discuss the discovery and scientific significance of AT 2021lwx ("Scary Barbie") for a broad public audience.

SELECTED MEDIA

- Why does a leap year have 366 days?, 2024, [Curious Kids](#), [The Conversation](#)
- Scientific Method: When a supermassive black hole devours a star, 2023, [NPR 1A Podcast](#)
- Astronomers detect "Scary Barbie" supermassive black hole ripping apart huge star in "terrifying" spaghettification event, 2023, [CBS](#)
- ‘Terrifying’: Why the universe’s largest cosmic explosion is called ‘Scary Barbie’, 2023, [USA Today](#)
- Uncovering a star’s demise: Supermassive black hole tears apart a giant star in a display brighter, more energetic and longer lasting than any observed before, 2023, [Purdue News](#)
- Master of Ceremony - JWST Launch Party, 2021, [YouTube](#)
- Saturday Morning Astrophysics Outreach, 2021, [YouTube](#)
- Stellarium Lesson Intro, 2021, [YouTube](#)

REFERENCES

Prof. David J. Sand

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